

ORIGINAL PAPER

Prolonged gut microbial alterations in post-transplant survivors of allogeneic haematopoietic stem cell transplantation

Akihisa Hino¹  | Kentaro Fukushima¹  | Shinsuke Kusakabe¹  | Tomoaki Ueda¹ |
 Takao Sudo^{1,2} | Jiro Fujita¹ | Daisuke Motooka^{2,3} | Aya K. Takeda⁴ |
 Natsuko O. Shinozaki⁴ | Satoshi Watanabe⁴ | Takafumi Yokota^{1,5} |
 Hirohiko Shibayama^{1,6} | Shota Nakamura^{2,3} | Naoki Hosen^{1,2,7}

¹Department of Hematology and Oncology, Osaka University Graduate School of Medicine, Suita, Japan

²Integrated Frontier Research for Medical Science Division, Institute for Open and Transdisciplinary Research Initiatives (OTRI), Osaka University, Suita, Japan

³Department of Infection Metagenomics, Research Institute for Microbial Diseases, Osaka University, Suita, Japan

⁴Cykinso, Inc., Tokyo, Japan

⁵Department of Hematology, Osaka International Cancer Institute, Osaka, Japan

⁶Department of Hematology, National Hospital Organization Osaka National Hospital, Osaka, Japan

⁷Laboratory of Cellular Immunotherapy, World Premier International Immunology Frontier Research Center, Osaka University, Suita, Japan

Correspondence

Kentaro Fukushima, Department of Hematology and Oncology, Osaka University Graduate School of Medicine, Suita, Osaka 565-0871, Japan.
 Email: kfukushi@bldon.med.osaka-u.ac.jp

Shota Nakamura, Department of Infection Metagenomics, Research Institute for Microbial Diseases, Osaka University, Suita, Osaka 565-0871, Japan.
 Email: nshotat@gen-info.osaka-u.ac.jp

Summary

Dysbiosis of the gut microbiota has been reported to increase early complications after allogeneic haematopoietic stem cell transplantation (allo-HSCT). However, it remains unclear whether gut microbial alterations persist during late complications, such as chronic graft-versus-host disease (cGVHD) or secondary cancers. Here, we analysed the gut microbiota of 59 patients who survived for 1–21.7 years (median, 6.4 years) after allo-HSCT. Long-term survivors showed lower gut microbial diversity than the age- and sex-matched healthy controls. This decreased diversity was reflected in the reduced abundance of the butyrate-producing bacteria. Patients with a history of grade 3 acute graft-versus-host disease (aGVHD) exhibited higher *Veillonella* abundance than patients with a history of grade 1–2 or non-aGVHD cases. The abundance of *Faecalibacterium* showed no decrease only in limited cGVHD cases. Additionally, the microbial structure in the secondary cancer group was significantly different ($p < 0.05$) from that in the non-secondary cancer group. This study is the first to show that microbial dysbiosis is present over a 10-year lifetime after discharge following allo-HSCT. Our results suggest that these prolonged gut microbial alterations may be associated with the development and exacerbation of late complications in post-transplant survivors.

KEYWORDS

aGVHD, allogeneic haematopoietic stem cell transplantation, cGVHD, gut microbiota, second cancer

INTRODUCTION

The outcomes of allogeneic haematopoietic stem cell transplantation (allo-HSCT) have improved with the advent of new therapies and drugs.¹ Such improvements have led to an increase in the number of long-term survivors. Late complications after transplantation are associated with a high mortality rate and decline in quality of life and

represent a major concern for long-term survivors. Chronic graft-versus-host disease (cGVHD) and secondary cancers are particularly concerning.^{2,3} cGVHD occurs in 54% of patients following transplantation and accounts for 37.8% of post-transplant, non-relapse mortality.^{4,5} The pathogenesis of cGVHD involves uncontrolled B-cell activation, immune activation of specific T-cell subsets, and a deficiency of regulatory T cells.⁶

Secondary cancer after allo-HSCT occurs at twice the rate found in the general population and accounts for 5%–10% of late deaths.^{7,8} Several risk factors for secondary cancers have been reported, including long-term immunosuppression, radiation therapy, chemotherapy, and tissue damage during cGVHD.⁸ The development of these major late complications is associated with post-transplant immune functions. However, immune functions in long-term post-transplant survivors have not been fully elucidated.

The human intestinal tract is a habitat for 40 trillion bacteria, collectively known as the gut microbiota. The gut microbiota plays an essential role in sustaining human life by regulating immunity, physiological functions, and nutrition.^{9,10} Abnormalities in the gut microbiota have been associated with various diseases, including carcinogenesis, autoimmune diseases, and infection.^{11,12} After allo-HSCT, the gut microbiota shows decreased microbial diversity and domination by a single bacterium, such as *Enterococcus* or *Streptococcus*.^{13,14} The decreased diversity has also been associated with increased risks for acute graft-versus-host disease (aGVHD) and aGVHD-related mortality.¹⁵ Patients are generally discharged within 1 to 3 months after allo-HSCT and gradually return to the same lifestyle as healthy individuals.^{16,17} A study that investigated gut microbial alterations both pre and post transplantation found that the decrease in microbial diversity observed after transplantation persists for up to 1 year after transplantation.¹⁸ However, whether long-term alterations in the gut microbiota persist for more than a year remains unexplored. Moreover, long-term survivors live with the risk of post-transplant late complications, but it is unknown how late complications affect the gut microbiota. Therefore, in the present study, we studied prolonged microbial alterations in post-transplant survivors and analysed the gut microbiota of 59 patients who survived for 1–21.7 years after allo-HSCT.

PATIENTS AND METHODS

Study design

The Institutional Review Board of Osaka University Hospital (Osaka, Japan) approved this study (approval number 16198), which was conducted in compliance with the Declaration of Helsinki (2013). All patients provided written informed consent for sample collection. We enrolled 84 patients who had undergone allo-HSCT at Osaka University Hospital between September 1997 and June 2018 and survived for at least 12 months. Patients ($n = 25$) with insufficient faecal samples during sample collection were excluded from our cohort. Finally, we collected 59 faecal samples (one sample from each patient). All patients included in the study were not hospitalised at the time of specimen collection and were released from dietary restrictions after 1 year post transplant. In addition, 59 faecal samples from healthy controls (HCs) were analysed. HCs were selected from 10 063 individuals registered in the Mykinso cohort¹⁹ from January

2017 to April 2020. From this population, we selected potential HCs ($n = 2202$) to include in this study that met the following inclusion criteria: no history of the disease under treatment, no history of enteritis within one month of sample collection, no ongoing medication at the time of sample collection, no history of antibiotic use within one month of sample collection, and no supplement use (all self-reported by questionnaire). Subsequently, 59 HCs matched for sex and age were selected through matched-pair analysis. All the HCs provided informed consent for enrolment. The recruitment of volunteer donors was conducted according to the principles of the Declaration of Helsinki (2013), approved by the Cykinso Research Ethics Committee (No. LD-001-04 and LD-002-03), and registered with the UMIN Clinical Trials Registry (no. UMIN000028887 and UMIN000028888).

Collection and storage of faecal samples

Faecal samples were collected in faeces tubes (76 × 20 mm, Sarstedt, Nümbrecht, Germany) by the participants themselves and dissolved in RNAlater™ Stabilisation Solution (Thermo Fisher Scientific, Waltham, MA, USA) at -80°C .

Bacterial DNA extraction and analysis of the gut microbiota

Bacterial DNA was extracted from faecal samples using the DNeasy PowerSoil Kit (Qiagen, Venlo, Netherlands). We sequenced the V1–V2 variable region of the 16S rRNA gene on the Illumina MiSeq platform (Illumina, San Diego, CA, USA) using 251-base pair paired-end sequencing. The paired-end sequences obtained were merged, filtered, and denoised using the DADA2 software (<https://benjjneb.github.io/dada2/>). Taxonomic assignments were performed using the QIIME2 feature-classifier plugin in the Greengenes database (13_8 release) (<https://greengenes.secondgenome.com/>). The QIIME2 pipeline, version 2020.2, was used as the bioinformatics environment for processing all relevant raw sequencing data (<https://qiime2.org>).²⁰

Statistical analyses

Based on the data obtained from these analyses, we compared the gut microbiota of HCs and long-term survivors of allo-HSCT. Alpha and beta diversities were used to compare the gut microbiota. Alpha diversity reflects the degree of diversity and refers to the number of species that can be detected in a microbial ecosystem. Beta diversity reflects structural differences and indicates differences in microbial communities in different environments. The alpha diversity was assessed using the Shannon index.²¹ The beta diversity was assessed by determining weighted unique fraction (UniFrac) distances.²² Comparing the Shannon index values among the groups and determining the relative abundance of each

bacterium was done by performing the Kruskal–Wallis test, followed by the Mann–Whitney *U*-test with Bonferroni's adjustment as a post-hoc test. Comparisons of weighted UniFrac distances were performed using permutational analysis of variance. The threshold for statistical significance was set at $p < 0.05$. Linear discriminant analysis effect size (LEfSe) was performed using the Galaxy web application (<https://usegalaxy.org/>).²³ Bacteria with abundances below 1% were not evaluated. For the LEfSe, the threshold of the logarithmic linear discriminant analysis (LDA) score was set to ± 3.0 .

RESULTS

Patient characteristics

In this study, we enrolled 59 patients who underwent allo-HSCT and survived for 1–21.7 years. Faecal specimens of these patients were collected at a median of 6.4 years after allo-HSCT. The study population comprised 23 female and 36 male patients aged 24–71 years (Table 1). Since the gut microbiota has been reported to be affected by age and sex,¹⁹ samples were collected from age- and sex-matched individuals (HCs). The clinical characteristics of the patients are shown in Table 2. Approximately half of these patients had a history of aGVHD ($n = 25$) or cGVHD ($n = 23$). Mild aGVHD cases (grade 1: 9; grade 2: 12) and severe aGVHD cases (grade 3: 4) were classified according to the modified Glucksberg criteria.²⁴ Gastrointestinal symptoms were present in all severe aGVHD cases. Severe aGVHD cases had a median time of 3.4 years (3–8.7 years) and mild GVHD cases had a median time of 5.6 years (1–13.4 years) after allo-HSCT. We identified nine cases of limited cGVHD and 14 cases of extensive cGVHD according to classic cGVHD diagnostic criteria.²⁵

TABLE 1 Characteristics of patients and healthy controls (HCs)

Baseline characteristics	Patients ($n = 59$)		Healthy controls ($n = 59$)	
	<i>n</i>	%	<i>n</i>	%
Sex				
Female	23	39	23	39
Male	36	61	36	61
Age at collect (year)				
Median (range)	54	(24–71)	50	(20–70)
20–29-year-olds	1	1.7	1	1.7
30–39-year-olds	8	13.6	8	13.6
40–49-year-olds	10	16.9	10	16.9
50–59-year-olds	21	35.6	21	35.6
60–69-year-olds	17	28.8	17	28.8
Over 70-year-olds	2	3.4	2	3.4

Note: Baseline characteristics (sex and age) at the time of sample collection from long-term survivors and HCs. The HCs were matched according to the age distribution and sex.

Figure 1 shows the post-transplant progress (left side) and gut microbiota composition (right side) of each patient. All patients were cured of aGVHD at the time of sample collection as aGVHD is a complication seen early in the post-transplant period. However, most patients with cGVHD (19/23) were still receiving treatment at the time of specimen collection. Twelve of these patients were treated with calcineurin inhibitors or glucocorticoids for extensive cGVHD. Seven patients received topical treatment (two patients with improving extensive cGVHD and five with limited cGVHD). Twenty-six patients had been taking antibiotics within six months. Twenty-one of these patients had received sulfamethoxazole-trimethoprim prophylaxis for *Pneumocystis jirovecii* pneumonia. These patients were still taking prophylactic treatment at the time of specimen collection. The other five patients had taken levofloxacin or azithromycin against transient bacterial infection for 1 week. Seven patients were receiving probiotics. Fifteen patients were taking proton pump inhibitors.

Changes in the gut microbiota of long-term survivors after allo-HSCT

We compared the gut microbiota of long-term survivors with that of the HCs (Figure S1). The alpha diversity in the gut microbiota of the long-term survivors was significantly lower ($p < 0.001$; Shannon index) than that of the HCs (Figure 2A). The beta diversity of the gut microbiota of the long-term survivors differed significantly ($p < 0.001$; weighted UniFrac distance) from that of the HCs (Figure 2B). LEfSe analysis revealed that the abundance of the *Bacteroidetes* phylum was significantly higher ($p < 0.001$) in the patient group, whereas that of the *Actinobacteria* and *Firmicutes* phyla were decreased (Figure 2C,D). Within the *Bacteroidetes* phylum, the abundance of the *Bacteroides* genus increased significantly ($p < 0.001$), whereas that of the *Prevotella* genus decreased. The abundances of the *Coprococcus* and *Faecalibacterium* genera in the *Firmicutes* phylum and those of the *Bifidobacterium* genus in the *Actinobacteria* phylum decreased (Figure 2E,F). The relative abundance of the *Enterococcus* genus was higher in long-term survivors than in the HC group, but the relative proportion was small (median: 0%, range: 0%–5.23%). No significant increase in the abundance of *Streptococcus* genus was observed (Figure S2).

We compared gut microbiotas in patients at less than 3 years, 3–10 years, and more than 10 years after transplantation to assess changes over time. Relative to the gut microbiotas in HCs, the patients in all three categories of survival periods showed decreases in alpha diversity and changes in beta diversity. However, no significant differences in the microbiotas were observed among the three survival periods (Figure 3A,B). The relative abundances of *Bacteroides*, *Prevotella*, *Coprococcus*, *Faecalibacterium*, and *Bifidobacterium* did not significantly differ among the three categories after transplantation. However, the abundance of *Coprococcus* was significantly lower in all categories than in the HC group (Figure 3C). These results indicate that

TABLE 2 Characteristics of patients. Baseline characteristics of the long-term survivors

Clinical characteristics	n	%
Collection time after SCT (year)		
Median (range)	6.4	(1–21)
Donor source		
Unrelated	44	
Unrelated bone marrow	39	66.1
Unrelated umbilical cord blood	5	8.5
Related	15	
Related bone marrow	8	13.6
Related peripheral blood stem cell	7	11.9
Primary disease		
Acute myeloid leukaemia	31	52.5
Acute lymphoblastic leukaemia	14	23.7
Myelodysplastic syndrome	8	13.6
Other	6	10.2
Antibiotic within 6 months		
Yes	26	44.1
No	33	55.9
Probiotics		
Yes	7	11.9
No	52	88.1
Proton pump inhibitor		
Yes	15	25.4
No	44	74.6
aGVHD		
Non aGVHD	34	57.6
aGVHD	25	
Grade 1	9	15.3
Grade 2	12	20.3
Grade 3	4	6.8
Grade 4	0	0
cGVHD		
Non cGVHD	36	61
cGVHD	23	
Limited	9	15.3
Extensive	14	23.7
Secondary cancer		
Non secondary cancer	52	88.1
Secondary cancer	7	
Colorectal cancer	2	3.4
Tongue cancer	1	1.7

TABLE 2 (Continued)

Clinical characteristics	n	%
Oesophageal cancer	2	3.4
PTLD	2	3.4
Relapse		
Yes	2	3.4
No	57	96.6
Immunosuppressant		
No immunosuppressant	40	67.8
Topical treatment	7	11
Calcineurin inhibitor only	3	5.1
Glucocorticoids only	5	8.5
Calcineurin inhibitors and Glucocorticoids	4	6.8
Complications		
No existing complications	51	86.4
Non-infectious pulmonary complications	4	6.8
Idiopathic thrombocytopenic purpura	1	1.7
Cytomegalovirus retinitis	1	1.7
Chronic renal failure	1	1.7
Femoral head necrosis	1	1.7

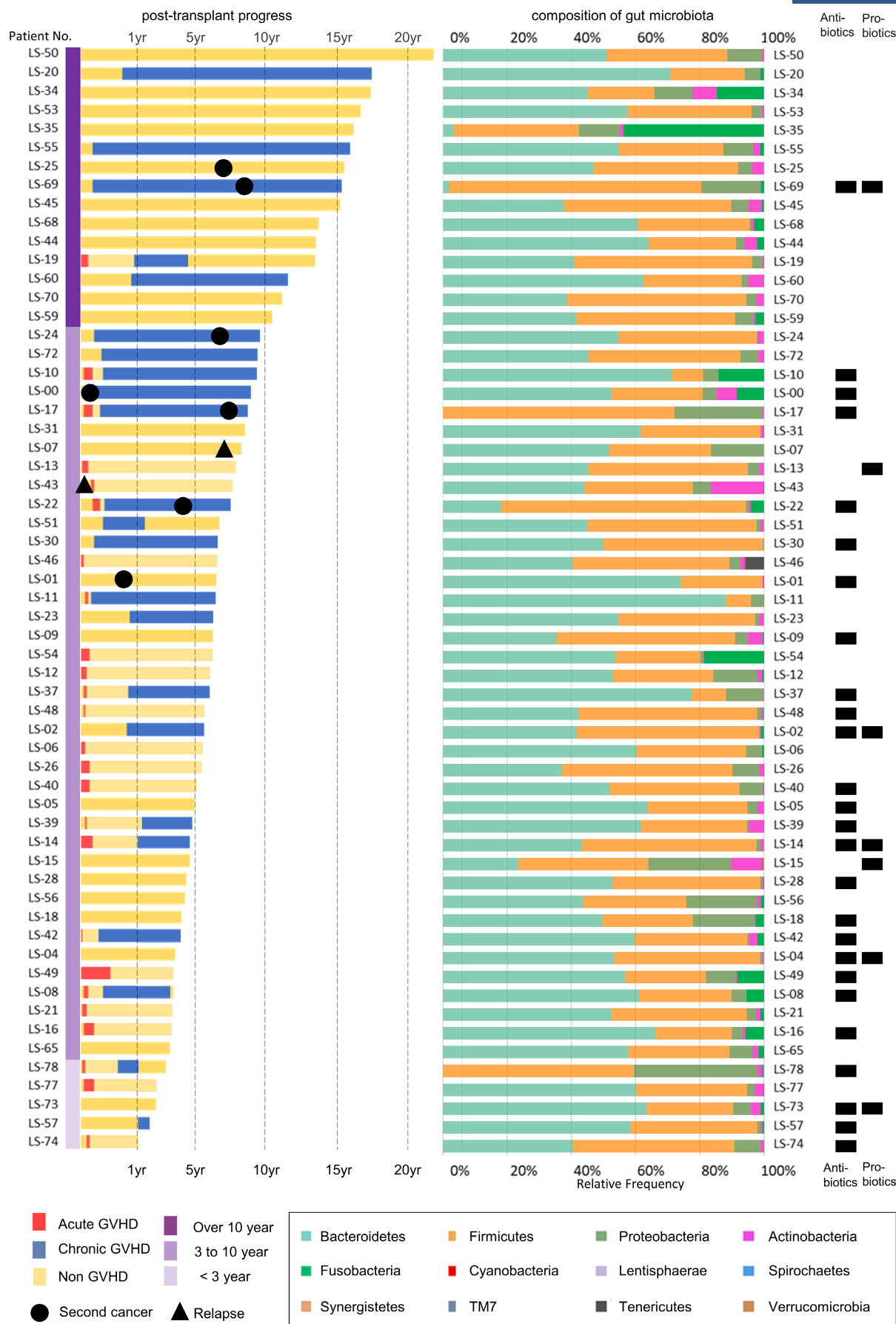
Note: Twenty-six patients had a history of antibiotic therapy within six months before sample collection, 17 who were taking trimethoprim/sulfamethoxazole as prophylaxis for *Pneumocystis jirovecii* pneumonia. The median onset of acute graft-versus-host disease (aGVHD) was 25 days (range: 6–88 days) post transplantation and that of chronic GVHD (cGVHD) was 157 days (range: 49–3089 days) post transplantation. Eleven patients developed both aGVHD and cGVHD. The median time before the onset of secondary cancer was 6.9 years post transplantation (range: 0.1–7.9 years), and the median time of specimen collection was 8.9 years post transplantation (range: 6.5–15.4 years).

alterations in the gut microbiota persisted for more than 10 years after transplantation, without improvements in diversity and structure. Age and sex did not affect the alpha and beta diversities of the gut microbiota (Figure S3A–D). In addition, the use of probiotics, antibiotics, and proton pump inhibitors (PPIs) did not affect the gut microbiota (Figure S4A–F).

Relationship between GVHD and the gut microbiota

We analysed relationships between the history of aGVHD and the gut microbiota of long-term survivors. Alpha diversity was not significantly different between patients who developed aGVHD and those who did not develop

FIGURE 1 Details of the post-transplantation course and gut microbiota in long-term survivors. The left side of the figure shows the post-transplantation course of each patient. The yellow, red, and blue areas show the durations that patients remained graft-versus-host disease (GVHD)-free, or experienced acute or chronic GVHD (aGVHD, cGVHD), respectively. The black circles indicate the onset of secondary cancer. The black triangles indicate recurrences. The right side of the figure shows the relative abundances of gut microbiota components at the phylum level. The data from patients who were taking antibiotics or probiotics are indicated with black bars. Patient numbers were assigned to 84 patients in the order of registration; however, 15 patients were not analysed due to insufficient sample collection.



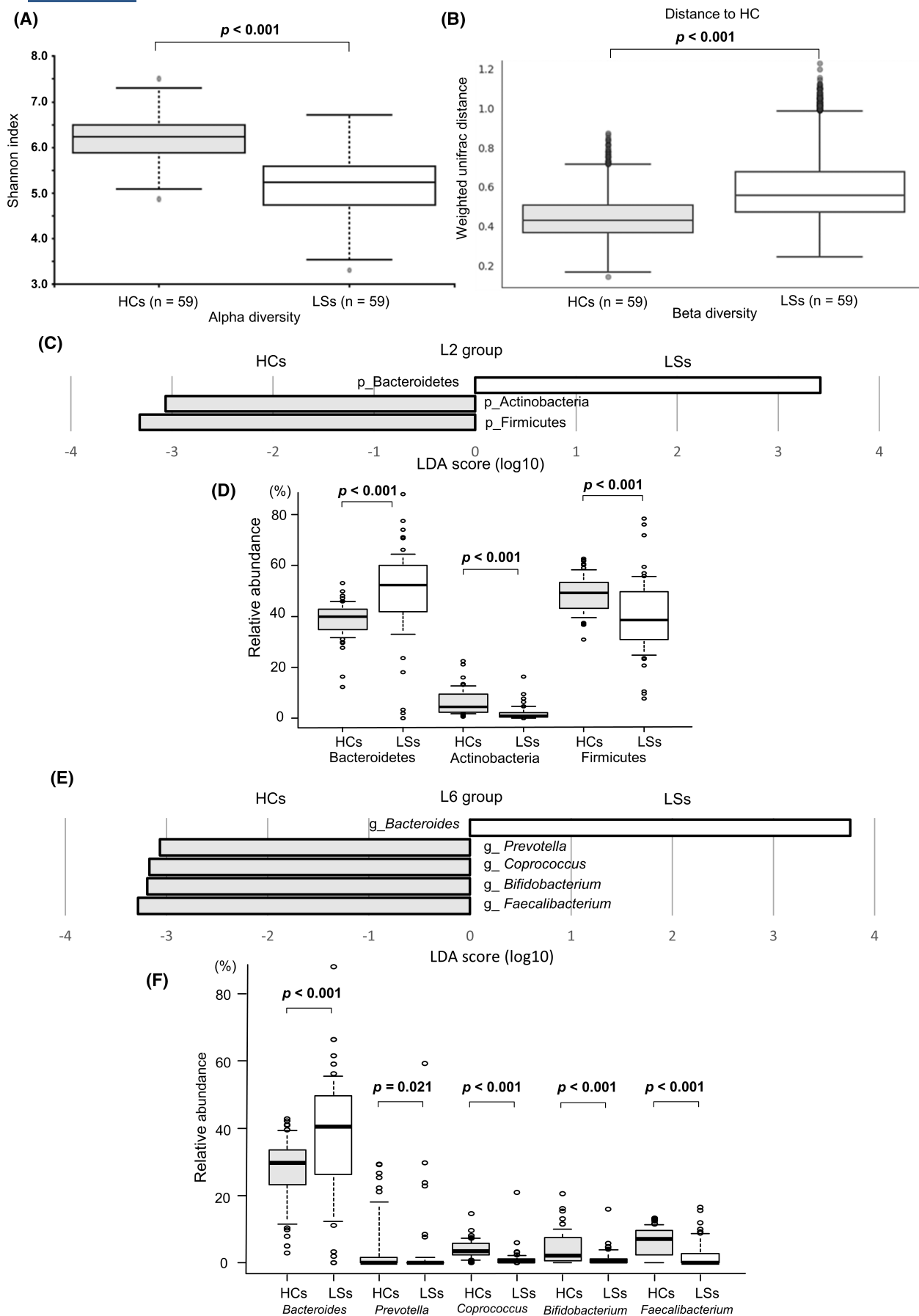


FIGURE 2 Gut microbiota diversities between the long-term survivors and healthy controls (HCs). (A,B) The alpha and beta diversities were compared between the long-term survivors and the HCs. (A) The box plots showing differences in alpha diversity were based on the Shannon index. (B) Differences in the beta diversity were based on differences reflected by the weighted UniFrac distances. (C) Linear discriminant analysis effect size (LEfSe) data for long-term survivors and HCs at the phylum level. The horizontal bars represent the effect size for each taxon in terms of $\log_{10}$ -transformed scores. Phyla with a p -value of <0.05 and a linear discriminant analysis (LDA) score of ≥ 3 or ≤ -3 were considered significant and are shown in the horizontal bar. (D) The box plot shows a comparison of the relative abundances at the phylum level. (E) LEfSe data at the genus level. Genera with a p -value of <0.05 and an LDA score of ≥ 3 or ≤ -3 were considered significant and are shown in the horizontal bar. (F) The box plot shows a comparison of the relative abundances at the genus level. Long-term survivors after allogeneic haematopoietic stem cell transplantation (allo-HSCT). (D,F) Significant p -values are shown in bold. LS, long-term survivor; p, phylum; g, genus.

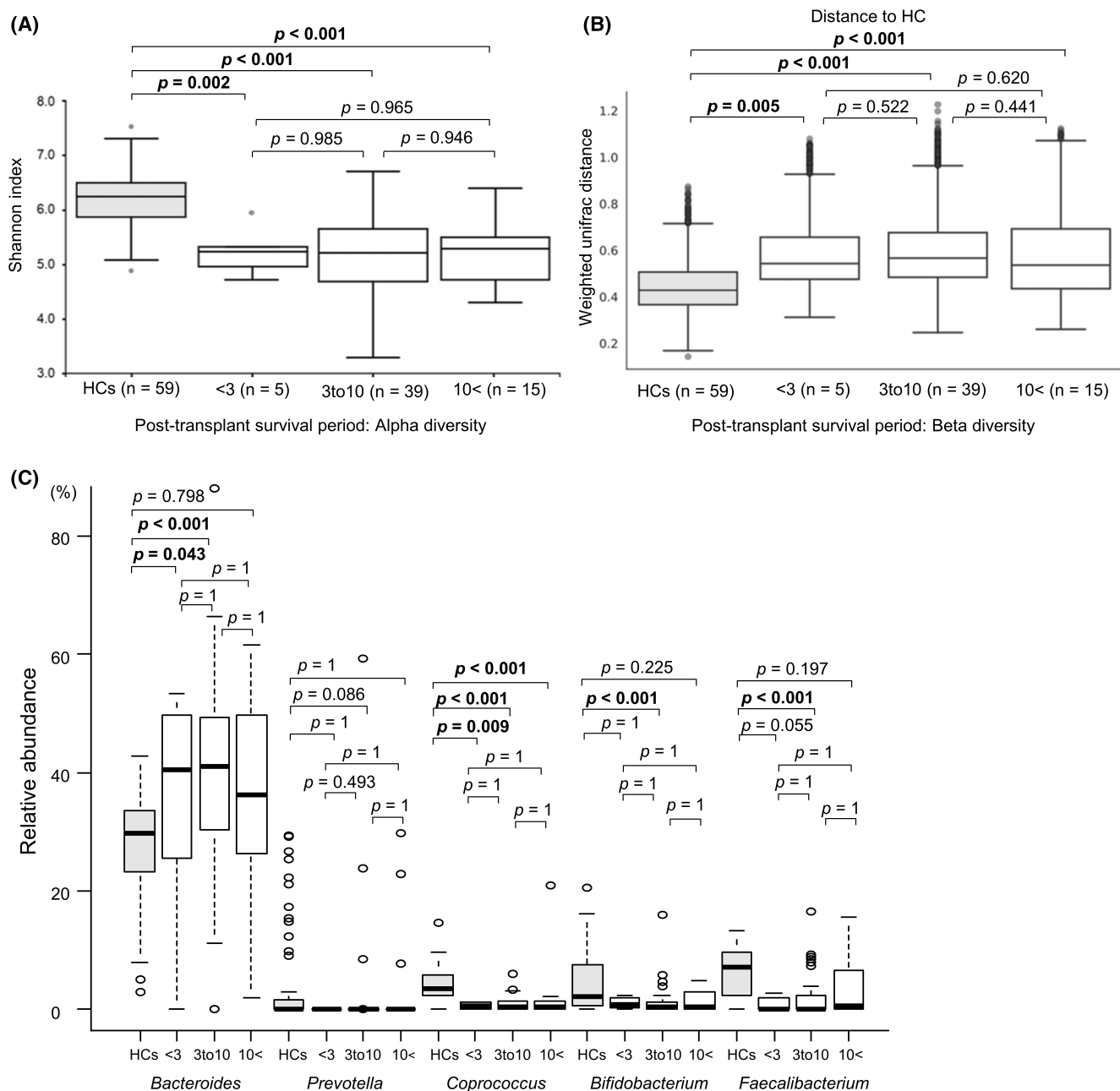


FIGURE 3 Comparison of the gut microbiota diversity according to the post-transplantation survival period. (A,B) The post-transplantation survival period was categorised into three groups (<3 years, 3–10 years, and >10 years), and the microbiota data for the patients in each group were compared with those of healthy controls (HCs). (A) Comparison of alpha diversities using the Shannon–Wiener index. (B) Comparison of beta diversities using weighted unique fraction (UniFrac) distances. (C) Relative abundances at the genus level were compared between HCs and patients at each post-transplantation survival period. (A–C) significant p -values are shown in bold.

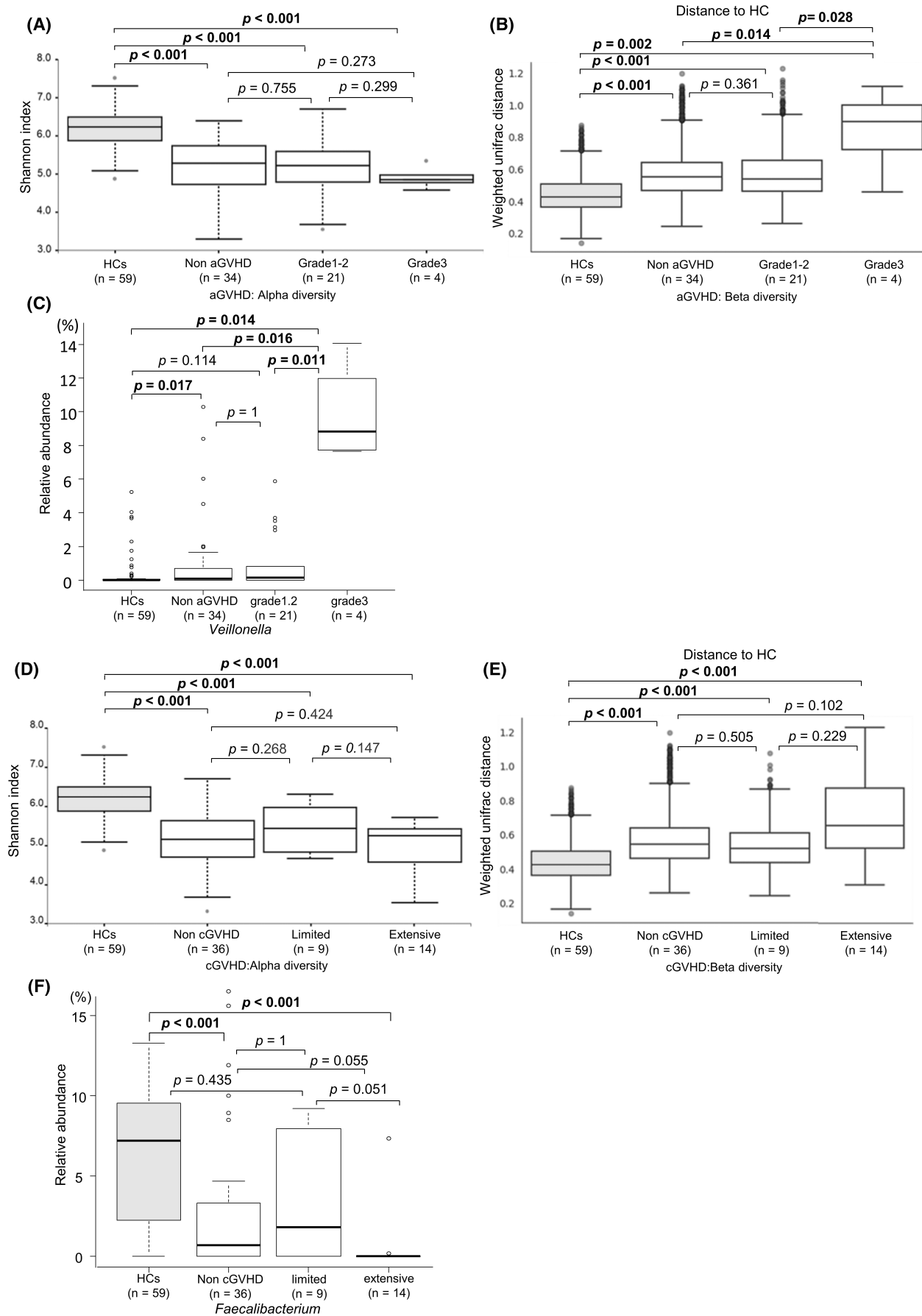


FIGURE 4 Comparison of the gut microbiota diversity according to the severity of acute and chronic graft-versus-host disease (aGVHD and cGVHD). (A,B) The alpha and beta diversities were compared between the non-acute GVHD, grade 1–2 aGVHD, grade 3 aGVHD, and healthy control (HC) groups. (A) Comparison of the alpha diversities using the Shannon–Wiener index. (B) Comparison of the beta diversities using weighted unique fraction (UniFrac) distances. (C) The relative abundances of *Veillonella* in patients with differing severities of aGVHD were compared to those in HCs. (D,E) The alpha and beta diversities were compared between the non-cGVHD, limited cGVHD, extensive cGVHD, and HC groups. (D) Comparison of the alpha diversities using the Shannon–Wiener index. (E) Comparison of the beta diversities using weighted UniFrac distances. (F) The relative abundances of *Faecalibacterium* in patients with differing severities of cGVHD were compared to those in HCs. (A–F) Significant *p*-values are shown in bold.

aGVHD (Figure 4A). The beta diversity of patients with a history of grade 3 aGVHD was significantly different from that of patients with a history of grade 1–2 aGVHD (Figure 4B). An increased abundance of the *Veillonella* genus caused a change in beta diversity in patients who developed grade 3 aGVHD (Figure 4C). The relative abundances of other bacteria in aGVHD cases are shown in Figure S5.

Severe aGVHD is one of the risk factors for developing cGVHD.²⁶ The 11 patients, including two patients with a history of grade 3 aGVHD, developed cGVHD after aGVHD in this study. These patients with acute and chronic GVHD showed no differences in the alpha and beta diversity of the gut microbiota compared to the patients with aGVHD only and cGVHD only (Figure S6).

Most cGVHD cases were still active at the time of specimen collection, and no differences were found in the alpha and beta diversities between patients with or without cGVHD (Figure 4D,E). However, the relative abundance of *Faecalibacterium* did not decrease in patients with limited cGVHD compared with that in HCs (Figure 4F). The extensive cGVHD cases did not show significant differences from the non- and limited cGVHD cases in terms of beta diversity in contrast to the grade 3 aGVHD cases, but their gut microbiota did show reduced *Faecalibacterium* abundance. The relative abundances of other bacteria in cGVHD cases are shown in Figure S7.

Relationship between the development of secondary cancers and the gut microbiota

Secondary cancer is an important late complication after allo-HSCT and cGVHD. We observed secondary cancers in seven patients, including those who developed colorectal cancer, oesophageal cancer, tongue cancer, or post-transplant lymphoproliferative disorder (PTLD). One patient with PTLD received chemotherapy. The other patient's condition improved with the reduction of immunosuppressive drugs. All patients with colorectal, oesophageal, and tongue cancer improved with curative resection. Specimens were collected from patients with a history of secondary cancer (*n* = 7) at a median of 8.9 years after transplantation and 6.2 years after treatment for secondary cancer. The gut microbiota of the group with secondary cancer was compared with that of the group without secondary cancer and the HC group. The group with secondary cancer showed similar alpha diversities but significantly different

beta diversities (Figure 5A,B). Three-dimensional principal coordinate analysis plots based on weighted UniFrac distances revealed that the microbiota structures in patients with a history of secondary cancer were significantly aberrant (Figure 5C). The relative abundance of the *Bacteroides* genus in the Bacteroidetes phylum was significantly lower in the group with secondary cancer compared to that in the group without secondary cancer (Figure 5D). These results indicate that patients who developed secondary cancer maintained a gut microbiota composition different from that of patients who did not develop secondary cancer over a long period.

DISCUSSION

In this study, lower microbial diversity presented in long-term transplant survivors for more than 10 years after transplantation. Taur et al.¹³ reported that reduced diversity continued until 35 days after transplantation. Markey et al.¹⁸ studied the microbiota 1 year after transplantation and found that reduced diversity occurred from an early period up to 365 days post-transplantation and speculated that the microbial diversity decreased due to hospitalisation, dietary restrictions, and exposure to antibiotics and other drugs. PPIs are one of the drugs used for long periods after allogeneic haematopoietic stem cell transplantation. They could also be responsible for affecting the gut microbiota of long-term survivors.²⁷ In this study, at the time of sample collection, the patients were not hospitalised or under any dietary restrictions. Additionally, the use of antibiotics, probiotics, and PPIs did not significantly impact the reduced microbiota diversity of long-term survivors. These findings suggest that non-hospitalisation, removing dietary restrictions, discontinuing antibiotics and PPIs, and administering probiotics are insufficient to re-establish gut microbial diversity in long-term survivors.

Previous data showed that during the early post-transplantation period, a single bacterial taxon (such as *Enterococcus* or *Streptococcus*) represented at least 30% of the relative abundance in the reduced microbiota.^{13,14} This dominance remained in some cases at 100 days after transplantation.¹⁸ Although we did not observe the dominance of *Enterococcus* or *Streptococcus* in long-term survivors, the relative abundance of *Enterococcus* was slightly higher than that in the HC group, suggesting that dominance by a single bacterial taxon in the early post-transplant period is likely transient.

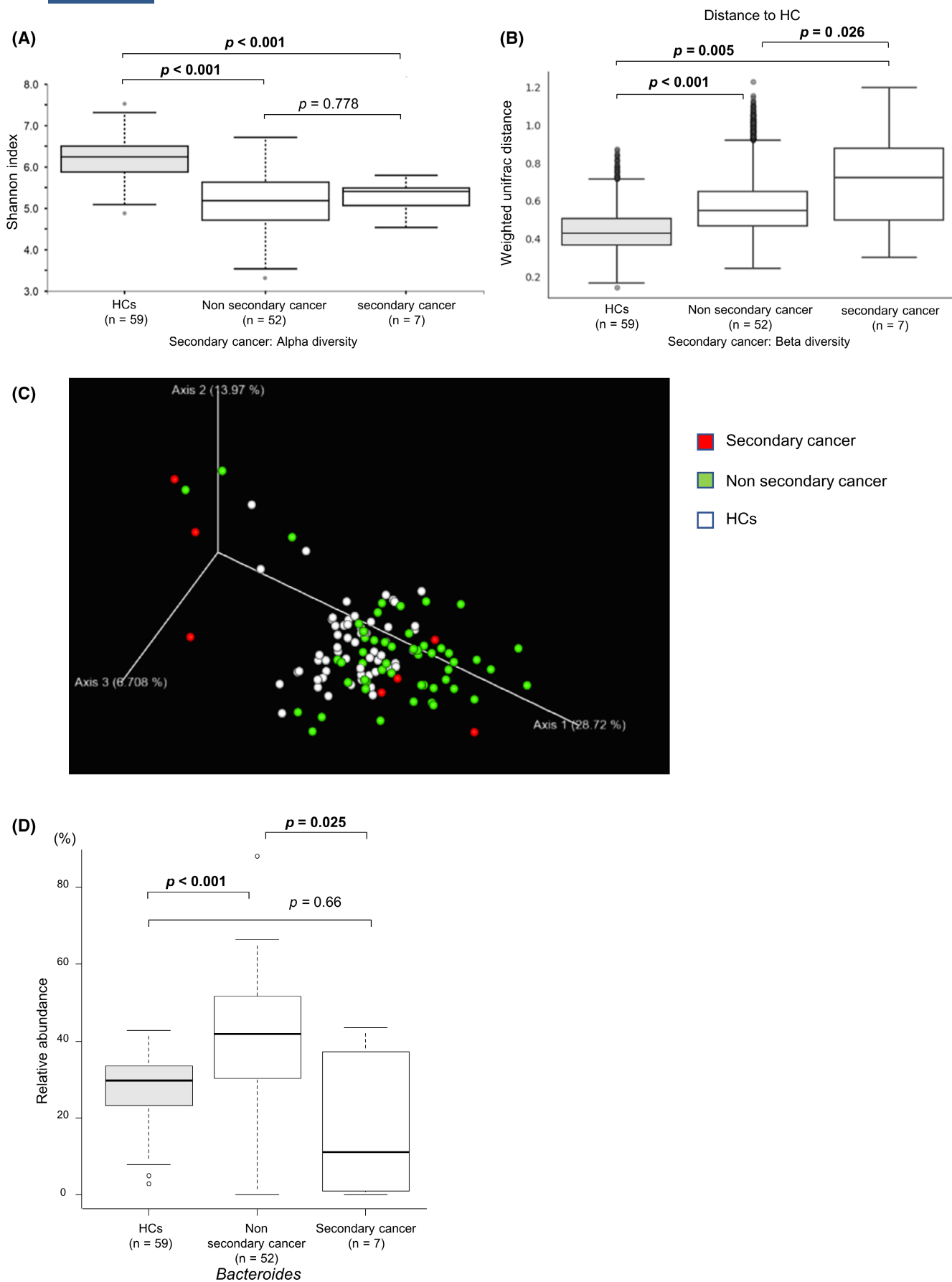


FIGURE 5 Comparison of gut microbiota diversities in secondary cancer. (A,B) The alpha and beta diversities were compared among patients with or without secondary cancer, as well as healthy controls (HCs). (A) Comparison of the alpha diversities using the Shannon–Wiener index. (B) Comparison of the beta diversities with weighted unique fraction (UniFrac) distances. (C) Three-dimensional principal coordinate analysis plots based on weighted UniFrac distances. Red, patients with secondary cancer ($n = 7$); green, patients without secondary cancer ($n = 51$); white, HCs ($n = 59$). (D) The relative abundances of the *Bacteroides* genus among patients in the three indicated groups. (A,B,D) Significant p -values are shown in bold.

We observed a difference in the microbial structure between long-term survivors and HCs. The abundance of *Bacteroides* increased in long-term survivors, whereas that of *Coprococcus*, *Faecalibacterium*, *Bifidobacterium*, and *Prevotella* decreased. Among them, *Coprococcus* and *Faecalibacterium* are butyrate-producing bacteria.²⁸ A decrease in the abundance of butyrate-producing bacteria 30 days after transplantation has been associated with increased mortality due to GVHD and transplantation.²⁹ Butyrate-producing bacteria can induce regulatory T-cell differentiation in the colon and suppress intestinal inflammation.³⁰ Owing to our retrospective study design, we cannot substantiate whether the decline in abundance of butyrate-producing bacteria was directly related to post-transplant complications. Based on the close relationship between the decrease in butyrate-producing bacteria and early post-transplant complications, we anticipated that the long-term decline in butyrate-producing bacteria would also be associated with the development of late complications.

Although patients with a history of grade 3 aGVHD showed the same alpha diversity as those with a history of grade 1–2 aGVHD, the structural diversity in patients with a history of grade 3 aGVHD differed significantly from that in patients with a history of grade 1–2 aGVHD or without aGVHD. In the microbiota of patients with a history of grade 3 aGVHD, we observed an increased *Veillonella* population. The genus *Veillonella* has been associated with increased aGVHD.^{31,32} Severe aGVHD causes intestinal mucosal destruction with multiple contiguous crypt dropouts.³³ Paneth cells in these crypts play essential roles in shaping the gut microbiota composition by secreting antimicrobial peptides, such as α -defensins.³⁴ A deficiency in α -defensin production by damaged Paneth cells was shown to decrease the gut microbiota diversity during aGVHD.³⁵ Previous studies and our results suggest that patients with severe aGVHD, which is associated with Paneth cell damage, show persistent long-term dysbiosis, as exemplified by the increased *Veillonella* population. Except for *Veillonella*, the abundance of *Blautia*, *Escherichia*, and other species was not significantly different in terms of severity of aGVHD in this study. The decrease in *Blautia* and increase in *Escherichia* are associated with aGVHD severity and mortality.^{35,36} The patients with these bacterial compositions at high risk of mortality may have died before this study was conducted. We suspect that this survivor bias affected the abundance of species that increased in long-term survivors.

cGVHD is associated with less severe intestinal mucosal damage than aGVHD. The severity of cGVHD did not affect the overall structural changes vs those observed in

patients with aGVHD. The relative abundance of almost all major bacteria did not change with cGVHD severity. Only *Faecalibacterium* showed minor community structural changes, with no effect on the overall structure. The abundance of *Faecalibacterium* decreased in the extensive cGVHD cases and was unaffected in the limited cGVHD cases compared with the corresponding abundances in the HC group. A previous report showed that the presence of butyrate-producing bacteria, such as *Lachnospirillum*, *Clostridium*, and *Faecalibacterium*, could reduce the incidence of cGVHD.¹⁸ We speculate that long-term survivors who lack butyrate-producing bacteria continue to experience extensive cGVHD.

A secondary cancer is a major late complication after transplantation and cGVHD. The median duration to the onset of secondary cancers (3.5–6.8 years post transplantation) is longer than that of cGVHD (7.4 months post transplantation).^{37–39} Long observation periods are necessary to follow the onset of secondary cancers. In this study, specimens were collected from patients with a history of secondary cancer at a median of 8.9 years post transplant and 6.2 years after treatment of secondary cancer. The secondary cancer group showed significantly different microbiota structures compared to the HC and non-secondary cancer groups. The abundance of the *Bacteroides* genus increased in the non-secondary cancer group but remained similar in the secondary cancer group, compared to that in the HC group. *Bacteroides fragilis* induces carcinogenesis in the colon by metalloproteinase toxins.⁴⁰ The abundance of *Bacteroides fragilis* has been reported to increase in colorectal cancer tissues, but our results did not support these previous reports.⁴¹ The sample size of our cross-sectional study was too limited to draw definitive conclusions. Previous epidemiological studies of secondary cancers have been conducted with large cohorts of 1000 or more individuals.^{42–44} However, many transplant cases are required to identify the specific bacteria associated with the late complication, including carcinogenesis, and longitudinal studies over 10 years are difficult to conduct. Nevertheless, the results of the present study provide novel and important clues to unravelling the gut microbiota of long-term survivors.

The gut microbiota of long-term post-transplant survivors showed maintained structural alterations, with reduced butyrate-producing bacteria, for over 10 years. This altered gut microbiota acquired stability that was unaffected by antibiotics. Furthermore, conventional therapies, such as probiotics, failed to improve the diversity of the gut microbiota. We speculate that these prolonged structural alterations were associated with late complications after transplantation. Structural alterations with increased butyrate-producing

bacteria may cause mild cGVHD, whereas specific structural alterations may increase the risk of developing late complications. We believe that modifying the microbial structure into a high butyrate-producing bacterial population is a promising therapeutic strategy for reducing the risk of late post-transplantation complications.

AUTHOR CONTRIBUTIONS

Akihisa Hino, Kentaro Fukushima, Shinsuke Kusakabe, and Shota Nakamura designed the study; Akihisa Hino performed the statistical analysis; Akihisa Hino, Kentaro Fukushima, Shinsuke Kusakabe, and Shota Nakamura contributed to the manuscript preparation; Daisuke Motooka and Shota Nakamura performed the metagenomic analyses; Jiro Fujita managed the clinical data; Aya K. Takeda, Natsuko O. Shinozaki, and Satoshi Watanabe performed matched-pair analyses with HCs. Takao Sudo, Tomoaki Ueda, Takafumi Yokota, Hirohiko Shibayama, and Naoki Hosen critically revised the manuscript. All authors approved the submitted and final version of the manuscript.

ACKNOWLEDGEMENTS

This study was supported by the Japan Society for the Promotion of Science KAKENHI (grant number 20 K17401). We thank Yutaka Takashima and Henyun Shih for providing technical assistance. We also thank Motoko Oda for assistance in data management. We would like to thank Editage (www.editage.jp) for English language editing.

CONFLICT OF INTEREST

The authors declare no conflicts of interest associated with this study.

DATA AVAILABILITY STATEMENT

For original data, please contact kfukushi@bldon.med.osaka-u.ac.jp

ORCID

Akihisa Hino  <https://orcid.org/0000-0002-7331-2713>
 Kentaro Fukushima  <https://orcid.org/0000-0002-8003-2584>
 Shinsuke Kusakabe  <https://orcid.org/0000-0002-9814-7611>

REFERENCES

- Granot N, Storb R. History of hematopoietic cell transplantation: challenges and progress. *Haematologica*. 2020;105(12):2716–29.
- Schoemans HM, Lee SJ, Ferrara JL, Wolff D, Levine JE, Schultz KR, et al. EBMT-NIH-CIBMTR task force position statement on standardized terminology & guidance for graft-versus-host disease assessment. *Bone Marrow Transplant*. 2018;53(11):1401–15.
- Tichelli A, Beohou E, Labopin M, Socié G, Rovó A, Badoglio M, et al. Evaluation of second solid cancers after hematopoietic stem cell transplantation in European patients. *JAMA Oncol*. 2019;5(2):229–35.
- DeFilipp Z, Alousi AM, Pidala JA, Carpenter PA, Onstad LE, Arai S, et al. Nonrelapse mortality among patients diagnosed with chronic GVHD: an updated analysis from the chronic GVHD consortium. *Blood Adv*. 2021;5(20):4278–84.
- Fraser CJ, Bhatia S, Ness K, Carter A, Francisco L, Arora M, et al. Impact of chronic graft-versus-host disease on the health status of hematopoietic cell transplantation survivors: a report from the bone marrow transplant survivor study. *Blood*. 2006;108(8):2867–73.
- Zeiser R, Blazar BR. Pathophysiology of chronic graft-versus-host disease and therapeutic targets. *N Engl J Med*. 2017;377(26):2565–79.
- Rizzo JD, Curtis RE, Socie G, Sobocinski KA, Gilbert E, Landgren O, et al. Solid cancers after allogeneic hematopoietic cell transplantation. *Blood*. 2009;113(5):1175–83.
- Danylesko I, Shimoni A. Second malignancies after hematopoietic stem cell transplantation. *Curr Treat Options Oncol*. 2018;19(2):9.
- Wu HJ, Wu E. The role of gut microbiota in immune homeostasis and autoimmunity. *Gut Microbes*. 2012;3(1):4–14.
- Backhed F, Ley RE, Sonnenburg JL, Peterson DA, Gordon JI. Host-bacterial mutualism in the human intestine. *Science*. 2005;307(5717):1915–20.
- Lazar V, Ditu LM, Pircalabioru GG, Gheorghe I, Curutiu C, Holban AM, et al. Aspects of gut microbiota and immune system interactions in infectious diseases, immunopathology, and cancer. *Front Immunol*. 2018;9:1830.
- Zheng D, Liwinski T, Elinav E. Interaction between microbiota and immunity in health and disease. *Cell Res*. 2020;30(6):492–506.
- Taur Y, Xavier JB, Lipuma L, Ubeda C, Goldberg J, Gbournie A, et al. Intestinal domination and the risk of bacteremia in patients undergoing allogeneic hematopoietic stem cell transplantation. *Clin Infect Dis*. 2012;55(7):905–14.
- Peled JU, Gomes ALC, Devlin SM, Littmann ER, Taur Y, Sung AD, et al. Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation. *N Engl J Med*. 2020;382(9):822–34.
- Taur Y, Jenq RR, Perales MA, Littmann ER, Morjaria S, Ling L, et al. The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. *Blood*. 2014;124(7):1174–82.
- Hamada R, Arai Y, Kondo T, Harada K, Murao M, Miyasaka J, et al. Higher exercise tolerance early after allogeneic hematopoietic stem cell transplantation is the predictive marker for higher probability of later social reintegration. *Sci Rep*. 2021;11(1):7190.
- Godara A, Siddiqui NS, Munigala S, Dhawan R, Kansagra AJ, Rapoport AP, et al. Length of stay and hospital costs for patients undergoing allogeneic stem-cell transplantation. *JCO Oncol Pract*. 2021;17(3):e355–68.
- Markey KA, Schluter J, Gomes ALC, Littmann ER, Pickard AJ, Taylor BP, et al. The microbe-derived short-chain fatty acids butyrate and propionate are associated with protection from chronic GVHD. *Blood*. 2020;136(1):130–6.
- Watanabe S, Kameoka S, Shinozaki NO, et al. A cross-sectional analysis from the Mykinso cohort study: establishing reference ranges for Japanese gut microbial indices. *Biosci Microbiota Food Health*. 2021;40(2):123–34.
- Bolyen E, Rideout JR, Dillon MR, Bokulich NA, Abnet CC, al-Ghalith GA, et al. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. *Nat Biotechnol*. 2019;37(8):852–7.
- Shannon CE. A mathematical theory of communication. *Bell Syst Tech J*. 1948;27(4):623–56.
- Lozupone CA, Hamady M, Kelley ST, Knight R. Quantitative and qualitative beta diversity measures lead to different insights into factors that structure microbial communities. *Appl Environ Microbiol*. 2007;73(5):1576–85.
- Segata N, Izard J, Waldron L, Gevers D, Miropolsky L, Garrett WS, et al. Metagenomic biomarker discovery and explanation. *Genome Biol*. 2011;12(6):R60.
- Przepiorka D, Weisdorf D, Martin P, Klingemann HG, Beatty P, Hows J, et al. 1994 consensus conference on acute GVHD grading. *Bone Marrow Transplant*. 1995;15(6):825–8.
- Shulman HM, Sullivan KM, Weiden PL, McDonald GB, Striker GE, Sale GE, et al. Chronic graft-versus-host syndrome in man. A long-term clinicopathologic study of 20 Seattle patients. *Am J Med*. 1980;69(2):204–17.

26. Grube M, Holler E, Weber D, Holler B, Herr W, Wolff D. Risk factors and outcome of chronic graft-versus-host disease after allogeneic stem cell transplantation-results from a single-center observational study. *Biol Blood Marrow Transplant*. 2016;22(10):1781–91.
27. Imhann F, Bonder MJ, Vich Vila A, Fu J, Mujagic Z, Vork L, et al. Proton pump inhibitors affect the gut microbiome. *Gut*. 2016;65(5):740–8.
28. Louis P, Flint HJ. Formation of propionate and butyrate by the human colonic microbiota. *Environ Microbiol*. 2017;19(1):29–41.
29. Meedt E, Hiergeist A, Gessner A, Dettmer K, Liebisch G, Ghimire S, et al. Prolonged suppression of butyrate-producing bacteria is associated with acute gastrointestinal graft-vs-host disease and transplantation-related mortality after allogeneic stem cell transplantation. *Clin Infect Dis*. 2022;74(4):614–21.
30. Furusawa Y, Obata Y, Fukuda S, Endo TA, Nakato G, Takahashi D, et al. Commensal microbe-derived butyrate induces the differentiation of colonic regulatory T cells. *Nature*. 2013;504(7480):446–50.
31. Beckman MF, Morton DS, Bahrani Mougeot F, Mougeot JC. Allogeneic stem cell transplant-associated acute graft versus host disease: a computational drug discovery text mining approach using oral and gut microbiome signatures. *Support Care Cancer*. 2021;29(4):1765–79.
32. Heidrich V, Bruno JS, Knebel FH, de Molla VC, Miranda-Silva W, Asprino PF, et al. Dental biofilm microbiota Dysbiosis is associated with the risk of acute graft-versus-host disease after allogeneic hematopoietic stem cell transplantation. *Front Immunol*. 2021;12:692225.
33. Naymagon S, Naymagon L, Wong SY, Ko HM, Renteria A, Levine J, et al. Acute graft-versus-host disease of the gut: considerations for the gastroenterologist. *Nat Rev Gastroenterol Hepatol*. 2017;14(12):711–26.
34. Salzman NH, Hung K, Haribhai D, Chu H, Karlsson-Sjöberg J, Amir E, et al. Enteric defensins are essential regulators of intestinal microbial ecology. *Nat Immunol*. 2010;11(1):76–83.
35. Eriguchi Y, Takashima S, Oka H, Shimoji S, Nakamura K, Uryu H, et al. Graft-versus-host disease disrupts intestinal microbial ecology by inhibiting Paneth cell production of alpha-defensins. *Blood*. 2012;120(1):223–31.
36. Jenq RR, Taur Y, Devlin SM, Ponce DM, Goldberg JD, Ahr KF, et al. Intestinal *Blautia* is associated with reduced death from graft-versus-host disease. *Biol Blood Marrow Transplant*. 2015;21(8):1373–83.
37. Arora M, Cutler CS, Jagasia MH, Pidala J, Chai X, Martin PJ, et al. Late acute and chronic graft-versus-host disease after allogeneic hematopoietic cell transplantation. *Biol Blood Marrow Transplant*. 2016;22(3):449–55.
38. Shimoni A, Shem-Tov N, Chetrit A, Volchek Y, Tallis E, Avigdor A, et al. Secondary malignancies after allogeneic stem-cell transplantation in the era of reduced-intensity conditioning; the incidence is not reduced. *Leukemia*. 2013;27(4):829–35.
39. Gallagher G, Forrest DL. Second solid cancers after allogeneic hematopoietic stem cell transplantation. *Cancer*. 2007;109(1):84–92.
40. Sears CL, Geis AL, Housseau F. *Bacteroides fragilis* subverts mucosal biology: from symbiont to colon carcinogenesis. *J Clin Invest*. 2014;124(10):4166–72.
41. Shariati A, Razavi S, Ghaznavi-Rad E, Jahanbin B, Akbari A, Norzaee S, et al. Association between colorectal cancer and *Fusobacterium nucleatum* and *Bacteroides fragilis* bacteria in Iranian patients: a preliminary study. *Infect Agent Cancer*. 2021;16(1):41.
42. Kolb HJ, Socie G, Duell T, Van Lint MT, Tichelli A, Apperley JF, et al. Malignant neoplasms in long-term survivors of bone marrow transplantation. Late effects working Party of the European Cooperative Group for blood and marrow transplantation and the European late effect project group. *Ann Intern Med*. 1999;131(10):738–44.
43. Yokota A, Ozawa S, Masanori T, Akiyama H, Ohshima K, Kanda Y, et al. Secondary solid tumors after allogeneic hematopoietic SCT in Japan. *Bone Marrow Transplant*. 2012;47(1):95–100.
44. Curtis RE, Rowlings PA, Deeg HJ, Shriner DA, Socié G, Travis LB, et al. Solid cancers after bone marrow transplantation. *N Engl J Med*. 1997;336(13):897–904.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Hino A, Fukushima K, Kusakabe S, Ueda T, Sudo T, Fujita J, et al. Prolonged gut microbial alterations in post-transplant survivors of allogeneic haematopoietic stem cell transplantation. *Br J Haematol*. 2023;201(4):725–737. <https://doi.org/10.1111/bjh.18574>